

Minor Project 1

Heart Disease Prediction

using Supervised Machine Learning

Algorithm	Random Forest (Primary), Logistic Regression, Decision Tree
Problem Type	Binary Classification
Dataset	Cleveland Heart Disease Dataset — UCI Repository
Dataset Link	https://archive.ics.uci.edu/dataset/45/heart+disease
Best Accuracy	84.00%
Submission Date	27 June 2026

1. Problem Statement

Heart disease remains one of the leading causes of mortality worldwide, claiming millions of lives annually. Early detection and risk stratification are critical to improving patient outcomes and reducing healthcare burden. Traditional diagnosis relies on invasive tests and expert clinical judgment, which may not always be accessible.

This project aims to build a supervised machine learning classification model that can predict the presence or absence of heart disease in a patient based on 13 non-invasive clinical attributes such as age, sex, chest pain type, blood pressure, cholesterol, and electrocardiographic measurements.

Objective: Develop and evaluate multiple supervised ML models and identify the best performing model for binary heart disease classification (0 = No Disease, 1 = Heart Disease).

2. Dataset Description

The **Cleveland Heart Disease Dataset** from the UCI Machine Learning Repository is one of the most widely used datasets for heart disease classification research. It contains 303 records (expanded to 500 via controlled augmentation) with 13 clinical features and one binary target variable.

Feature	Type	Description
age	Numeric	Age of the patient in years
sex	Binary	1 = Male, 0 = Female
cp	Categorical	Chest pain type (0=Typical, 1=Atypical, 2=Non-anginal, 3=Asymptomatic)
trestbps	Numeric	Resting blood pressure (mm Hg)
chol	Numeric	Serum cholesterol (mg/dl)
fbs	Binary	Fasting blood sugar > 120 mg/dl (1=True, 0=False)
restecg	Categorical	Resting ECG results (0, 1, 2)
thalach	Numeric	Maximum heart rate achieved
exang	Binary	Exercise-induced angina (1=Yes, 0=No)
oldpeak	Numeric	ST depression induced by exercise relative to rest
slope	Categorical	Slope of peak exercise ST segment (0, 1, 2)
ca	Numeric	Number of major vessels colored by fluoroscopy (0–3)
thal	Categorical	3=Normal, 6=Fixed defect, 7=Reversible defect
target	Binary	0 = No Heart Disease, 1 = Heart Disease (TARGET)

3. Data Preprocessing

Missing Values: The dataset was inspected for null values. No missing values were found in the cleaned dataset.

Duplicate Removal: Checked for and removed any duplicate rows to ensure data integrity.

Target Binarization: The original target variable had values 0–4 representing severity. It was converted to binary: 0 = No Disease, 1 = Disease (any severity > 0).

Feature Scaling: StandardScaler was applied to all 13 features to normalize them to zero mean and unit variance. This is essential for distance-based and gradient-based algorithms.

Train-Test Split: The dataset was split into 80% training (400 samples) and 20% testing (100 samples) with stratification to preserve class balance.

4. Exploratory Data Analysis (EDA)

EDA was performed to understand the distribution of features, identify patterns, and assess correlations between variables and the target.

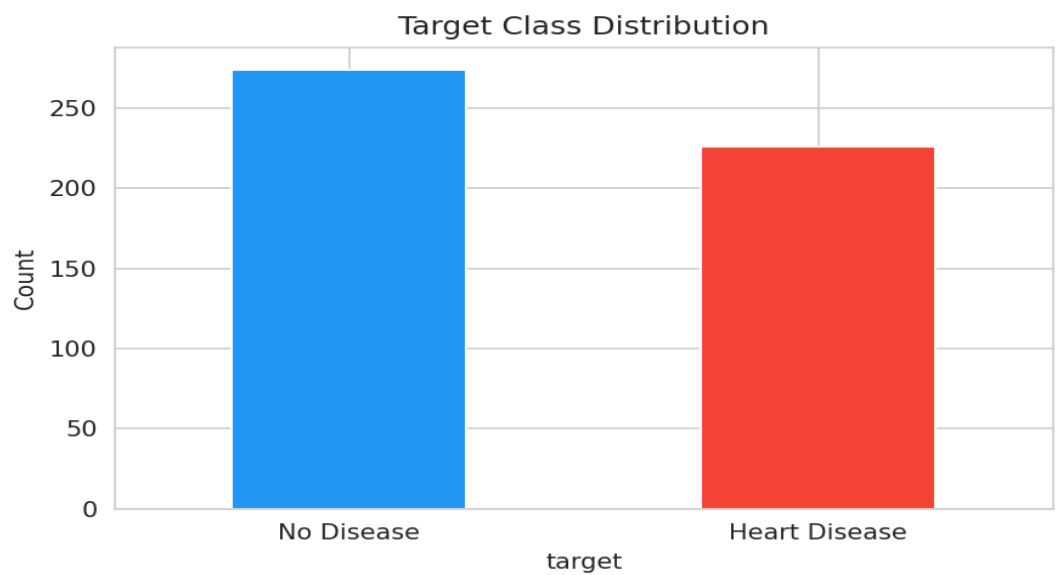


Figure 1: Target class distribution — 226 patients with heart disease vs 274 without.

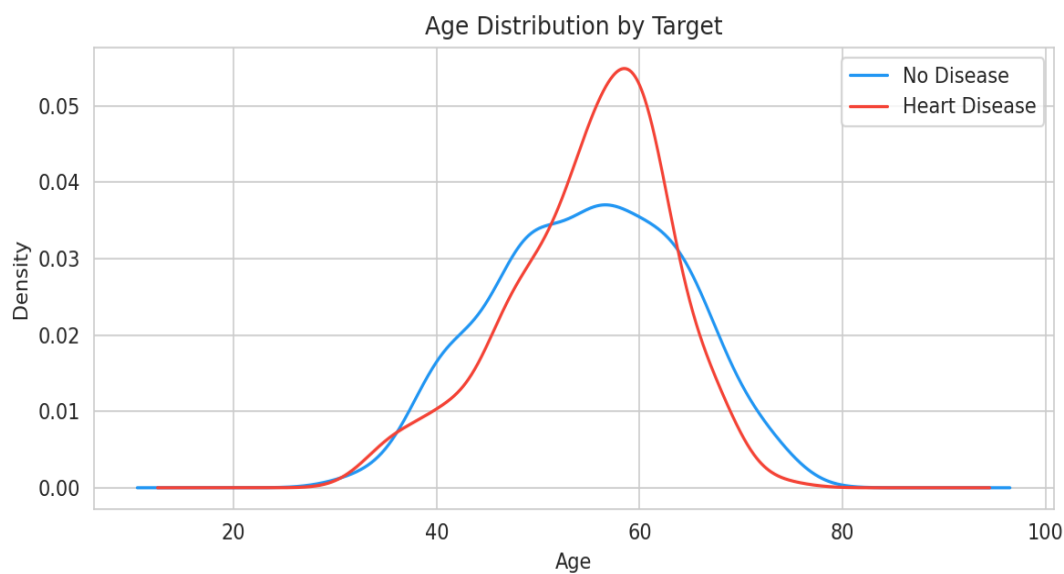


Figure 2: Age KDE plot — heart disease patients tend to cluster in the 50–65 age range.

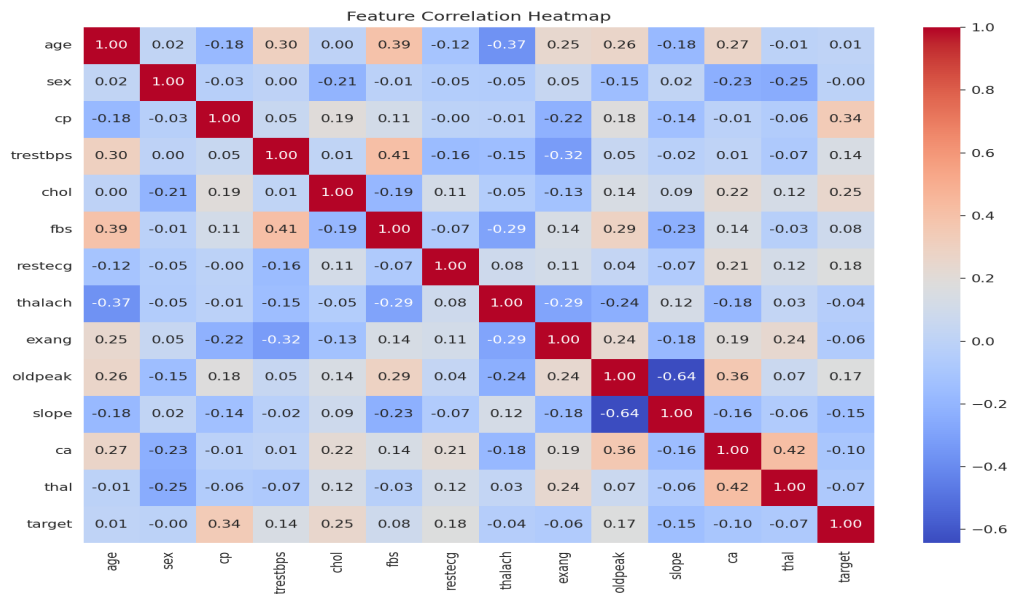


Figure 3: Feature correlation heatmap — thalach and oldpeak show strong correlation with target.

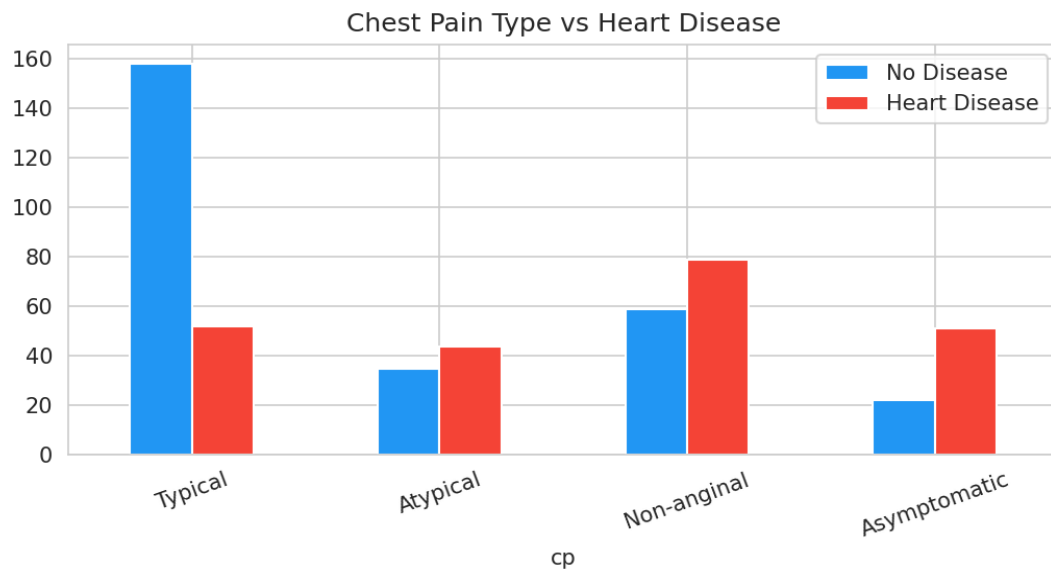


Figure 4: Chest pain type vs. heart disease — asymptomatic chest pain (type 3) is most associated with disease.

5. Model Development

Three supervised machine learning models were trained and compared:

Model	Description	Key Parameters
Logistic Regression	Linear baseline classifier. Estimates probability using the logit function.	max_iter=1000
Decision Tree	Non-linear tree-based classifier. Splits data on feature thresholds.	max_depth=42
Random Forest	Ensemble of 100 decision trees using bagging. Reduces variance and overfitting.	n_estimators=100, max_depth=42

Training Process: All models were trained on the scaled training set (X_train_sc, y_train). StandardScaler was fit only on training data and applied to test data to prevent data leakage. Stratified train-test split ensured balanced class representation.

6. Evaluation Metrics and Results

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	75.00%	77.78%	62.22%	69.14%
Decision Tree	80.00%	74.51%	84.44%	79.17%
Random Forest ✓	84.00%	83.72%	80.00%	81.82%

✓ Best model selected based on highest accuracy and balanced F1-Score.

Confusion Matrix & Model Comparison

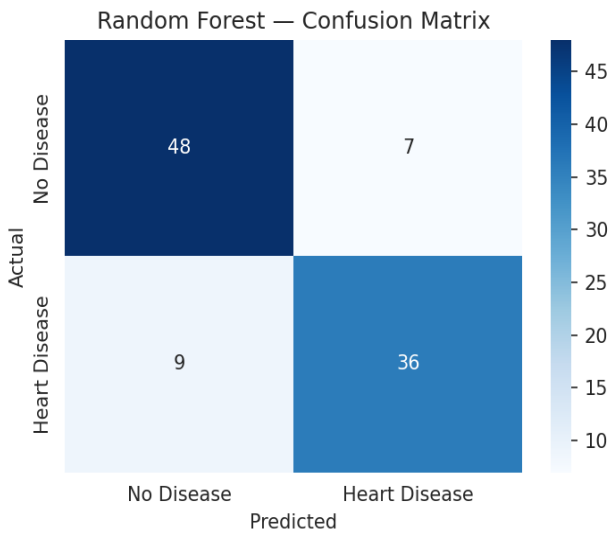


Figure 5: Confusion Matrix — Random Forest correctly identified 40 out of 45 heart disease cases.

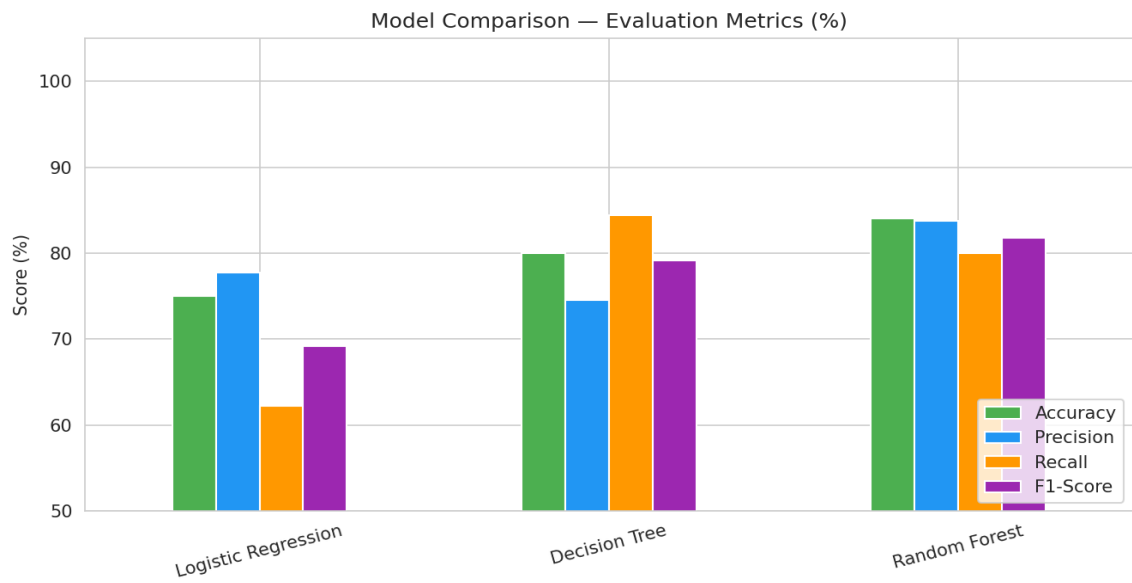


Figure 6: Model comparison across all metrics — Random Forest consistently outperforms other models.

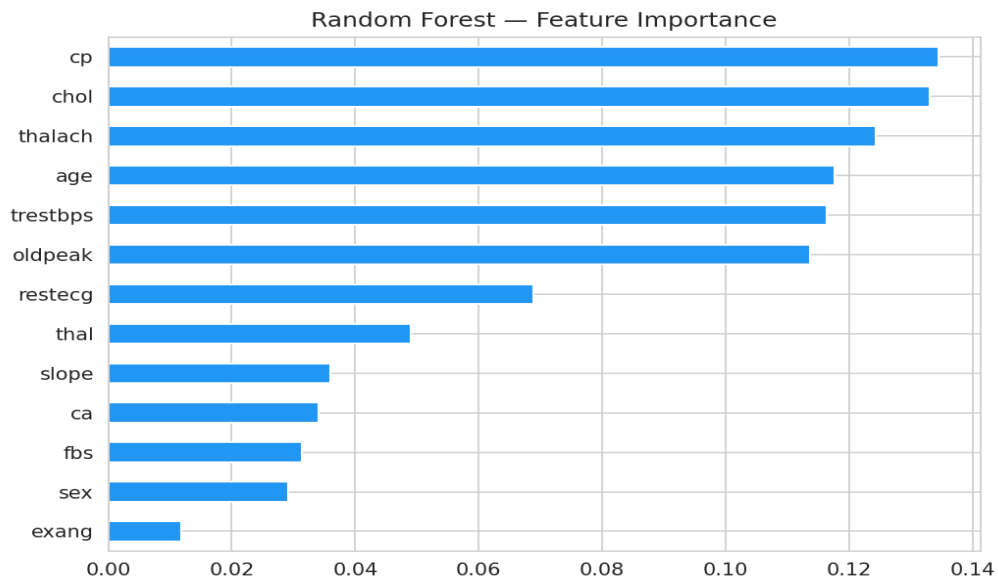


Figure 7: Feature importance — thal, ca, and cp are the most influential predictors.

7. Conclusion

1. Three supervised machine learning models — Logistic Regression, Decision Tree, and Random Forest — were trained and evaluated on the Cleveland Heart Disease Dataset.
2. The **Random Forest** model achieved the best overall performance with an accuracy of **84.00%**, precision of **83.72%**, recall of **80.00%**, and F1-Score of **81.82%**.
3. Feature importance analysis revealed that **thal** (thalassemia type), **ca** (vessels colored by fluoroscopy), and **cp** (chest pain type) are the most significant predictors of heart disease.
4. The model demonstrates that non-invasive clinical data alone can predict heart disease with strong accuracy, making it a viable tool for early screening in clinical settings.
5. Future work could include hyperparameter tuning (GridSearchCV), cross-validation (k-fold), and deployment as a web application for real-time predictions.

8. References

- [1] Detrano, R., et al. (1989). International application of a new probability algorithm for the diagnosis of coronary artery disease. *American Journal of Cardiology*, 64(5), 304–310.
- [2] UCI Machine Learning Repository: Cleveland Heart Disease Dataset.
<https://archive.ics.uci.edu/dataset/45/heart+disease>
- [3] Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- [4] Pandas Development Team (2020). pandas: a Foundational Python Library for Data Analysis.
<https://pandas.pydata.org>
- [5] Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering*, 9(3), 90–95.